Module 2: Virtual Machines and Virtualization of Clusters and Data Centers

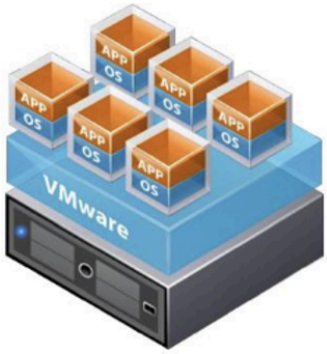
1. Introduction to Virtualization and Virtual Machines

What is Virtualization?

- Definition: Virtualization is the use of software methods to allow multiple virtual computing resources to run on a single physical hardware platform.
- Explanation: It involves creating a software-based (or virtual) representation of something, such as servers, storage, networks, or other physical resources. This allows a single physical machine to act as if it were multiple, separate machines.

What is a Virtual Machine (VM)?

- Definition: A VM is a software-based virtualization of a computer hardware system. It executes programs like a physical computer and can be interacted with in the same way.
- Explanation: A VM creates an isolated environment on a host machine that runs its own operating system (OS) and applications, behaving just like a separate physical computer.
- Key Points:
    - Allows multiple, different operating systems to run simultaneously on a single physical computer.
    - A VM is fundamentally a discrete set of files stored on a physical machine.



Main Files of a Virtual Machine

- Configuration File (`.vmx` / `.vmtx`):
  - Definition: The `.vmx` file is a plain-text file that serves as the blueprint or structural definition for the VM.
  - Explanation: It contains the current configuration and hardware settings for the VM. This includes details like the amount of RAM, network interface card (NIC) settings, CD-ROM connections, and parallel/serial port information. It also defines advanced resource and power settings.
  - Templates: When a VM is converted into a template (a master copy for creating new VMs), the `.vmx` file is replaced by a `.vmtx` file.
  - The .vmx file can be copied from the datastore and opened using a program that supports creation and saving of files using UTF-8 encoding, such as WordPad.
- Virtual Disk File:
  - Definition: This file stores the contents of the VM's hard drive.
  - Explanation: It contains the guest operating system, applications, and all user data. It appears to the guest OS as a standard physical hard drive.
- NVRAM Settings File (`.nvram`):
  - Definition: This is a small file that contains the BIOS settings for the VM.

    - Explanation: Just as a physical server has a BIOS chip for hardware configuration, the VM uses this file. The virtual BIOS settings can be accessed by pressing a key (e.g., F2) when the VM is powered on.
- Log File:
    - Definition: This file records the activity of the VM.
    - Explanation: It is used for troubleshooting and keeping a history of VM events and errors.

Purpose and Benefits of Virtualization

- Purpose: The primary purpose is to <u>enhance resource sharing among multiple users and improve computer performance</u> in terms of resource utilization and application flexibility.
- Explanation: By abstracting hardware resources, a single physical machine can be divided into multiple, efficient virtual environments.
- Key Points:
    - Both hardware resources (CPU, memory, I/O devices) and software resources (operating systems, software libraries) can be virtualized.
    - Benefit Example (Resource Utilization): Consider a server with a 20-core processor.
        - Without Virtualization: All 20 cores might be dedicated to a single application, leaving them underutilized if the application doesn't need that much power.
        - With Virtualization: These 20 cores can be split. For example, 3 cores to Application A1, 3 to Application A2, 4 to Application A3, leaving 10 cores for the host machine. This ensures a more effective and balanced utilization of the available resources.
- Virtualization as a Software-Defined Technology: Virtualization is the foundation for software-defined data centers, enabling:
    - Software-defined compute: Abstracting compute resources (CPU, RAM).
    - Software-defined networking: Abstracting network resources.
    - Software-defined storage: Abstracting storage resources.

2. Virtual Machine Architecture and Components

Core Components

- Virtual Machine (VM): A representation of a real machine using software that provides an operating environment capable of running a guest OS.
- Guest Operating System (Guest OS): An operating system (like Linux or Windows) that is installed and runs on a virtual machine. It behaves as if it were running on dedicated physical hardware.
- Virtual Machine Monitor (VMM) / Hypervisor: The virtualization layer that sits between the VM and the underlying hardware (the host platform). It is responsible for managing and isolating VMs.

Virtual Machine Architectures

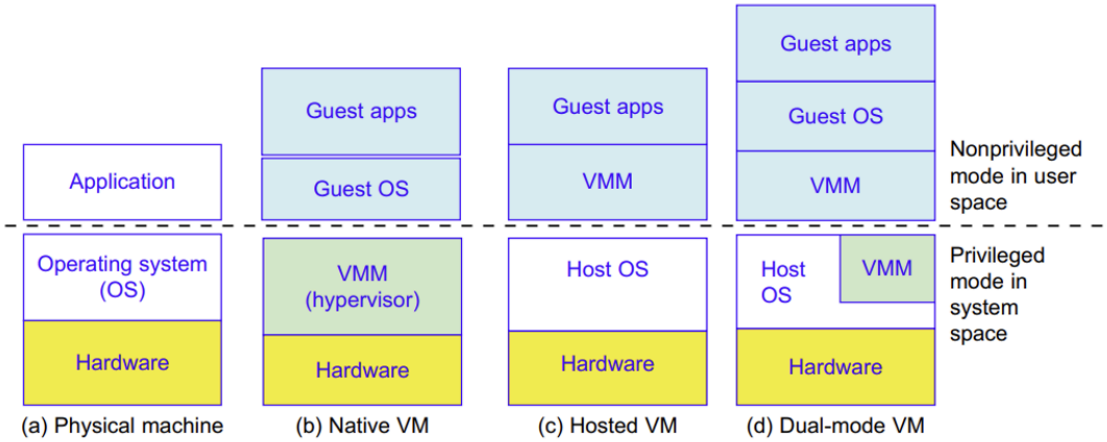


FIGURE 1.12

There are three main types of VM architecture, compared to a traditional physical machine.

- (a) Physical Machine:

    - Description: A standard computer where the host machine is equipped with physical hardware. The operating system runs directly on this hardware.
    - Example: An x86 architecture desktop computer running its installed Windows OS directly on the CPU and memory.
- (b) Native VM (Bare-Metal) Approach:
    - Description: A VMM (hypervisor) is installed directly onto the physical hardware. It runs in the most privileged mode and handles the CPU, memory, and I/O directly. Guest OSes then run on top of the hypervisor.
    - Key Feature: Also known as a Type 1 Hypervisor.
    - Example: The XEN hypervisor running directly on hardware, with a Linux system as a guest OS.
- (c) Hosted VM Approach:
    - Description: The VMM runs as an application on top of an existing host operating system. The VMM runs in a non-privileged mode of the host OS.
    - Key Feature: The host OS does not need to be modified.
    - Example: Running VMware Workstation on top of Windows.
- (d) Dual-Mode VM Approach:
    - Description: Part of the VMM runs at the user level, and another part runs at the supervisor (kernel) level.
    - Key Feature: To enable this, the host OS may need to be modified to some extent. This is a hybrid approach.

Primitive Operations of VMs

VMs are managed through several key operations:

- Multiplexing: Multiple VMs can be run and shared on a single hardware machine.
- Suspend/Resume: A running VM can be suspended, and its current state saved to stable storage (like a disk). It can later be resumed from that exact state.

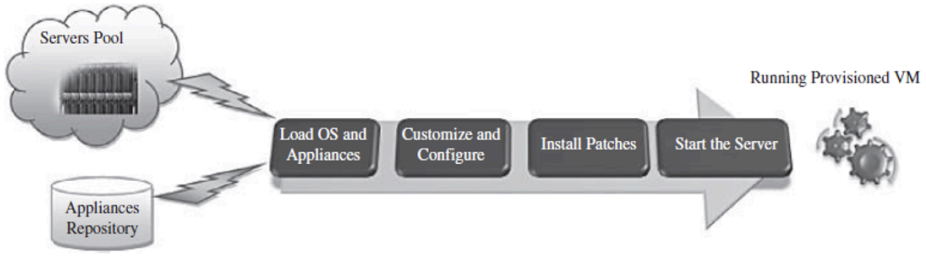
    - Explanation: Suspending saves the current state of all running applications. When resumed, everything continues as if it were never paused.
- Provisioning: The process of creating and preparing a new VM for use.
- Migration: A VM can be moved from one physical hardware platform to another.

3. Provisioning and Migration of Virtual Machines

Provisioning a Virtual Machine

The steps to provision a VM are as follows:

1. Select Server and OS Template: Choose a physical server from a pool of available resources. Then, select the appropriate operating system template for the new VM.
2. Load Software: Install the chosen OS, necessary device drivers, middleware, and any applications required for the service.
3. Customize and Configure: Configure the machine's network settings (e.g., IP address, gateway) and associate it with the required network and storage resources.
4. Ready to Start: The VM is now ready to be powered on and used.
- Example: In Microsoft Azure, this involves first selecting a server size and then choosing an OS image from a list (e.g., Ubuntu Server, Windows Server).



Live Migration (Hot Migration)

- Definition: The movement of a running virtual machine from one physical host to another with no perceived downtime. It is transparent to the end-users.
- When is it used?
    - Planned Maintenance: To move VMs off a host that requires hardware or software updates.
    - Load Balancing: To distribute VMs across hosts to optimize resource utilization (e.g., moving a VM from an overloaded host to a less busy one).
    - Server Consolidation: To pack VMs onto fewer hosts during periods of low activity to save power.
- Key Characteristic: When performed correctly, the process takes only milliseconds and is unnoticeable to end-users.

The 5 Stages of Live Migration

1. Stage 0: Pre-Migration: An active VM is running on physical Host A. The system determines which VM to migrate and selects a destination physical Host B.
2. Stage 1: Reservation: A request is made to migrate the VM from Host A to Host B. Host B checks if it has the necessary resources (CPU, memory) available to host the VM. If yes, it reserves a container of that size.
3. Stage 2: Iterative Pre-Copy: The entire execution state of the VM (primarily its memory) must be transferred to Host B.
    - First Iteration: All memory pages are copied from Host A to Host B.
    - Subsequent Iterations: While the first copy happens, the VM on Host A continues to run, and some memory pages may change (become "dirty"). Subsequent iterations only copy the pages that were dirtied during the previous transfer phase. This process repeats, reducing the number of dirty pages each time.
4. Stage 3: Stop-and-Copy:

    - The running OS instance on Host A is suspended. Applications stop running. This period of unavailability is called "downtime."
    - Network traffic is redirected to Host B.
    - The remaining inconsistent memory pages, along with the CPU state and network state, are transferred to Host B.
    - At the end of this stage, both Host A and Host B have a consistent, suspended copy of the VM. The copy on Host A is kept as the primary in case of failure.
5. Stage 4: Commitment:
    - Host B signals to Host A that it has successfully received a complete and consistent OS image.
    - Host A acknowledges this message, committing to the migration.
    - Host A can now discard the original VM. Host B becomes the new primary host.
6. Stage 5: Activation:
    - The migrated VM on Host B is activated. Post-migration code runs to reattach device drivers to the new machine.
    - The network connection is fully redirected to the new VM.
    - The VM reloads its states and resumes program execution. Service continues.

Live Migration Performance

- Key Principle: At least one host (either the source or destination) always has a consistent VM image during the entire process, ensuring data integrity.
- Experimental Result: A study showed the low overhead of live migration.
    - Throughput dropped only briefly during the precopy stages (e.g., from 870 MB/s to 694 MB/s) and experienced a very short downtime of just 165 ms. This minimal impact is critical for dynamic data center management and disaster recovery.

Regular/Cold Migration

- Definition: The migration of a virtual machine that is powered off.
- Key Differences from Live Migration:
    1. Shared Storage: Cold migration does not require the virtual machines to be on shared storage. Disks can be moved from one datastore to another. Live migration *does* require shared storage.
    2. CPU Checks: In live migration, strict CPU compatibility checks are required between source and destination hosts. These checks do not apply to cold migration, as the VM is not running.
- Process:
    1. Configuration files (`.vmx`, `.nvram`), log files, and virtual disks are moved from the source host's storage to the destination host's storage.
    2. The virtual machine is registered with the new host.
    3. The old version of the VM is deleted from the source host.

Applications of Migration

- Load Balancing: Moving VMs from overloaded hosts to less busy ones to ensure optimal performance. This can also make use of newly added server capacity.
- Failure Recovery: If a host fails, VMs can be restarted automatically on a different, healthy host.
- Maintenance: During planned maintenance, VMs are moved off a host before it is shut down, allowing for hardware or software updates without service disruption.

4. Levels of Virtualization Implementation

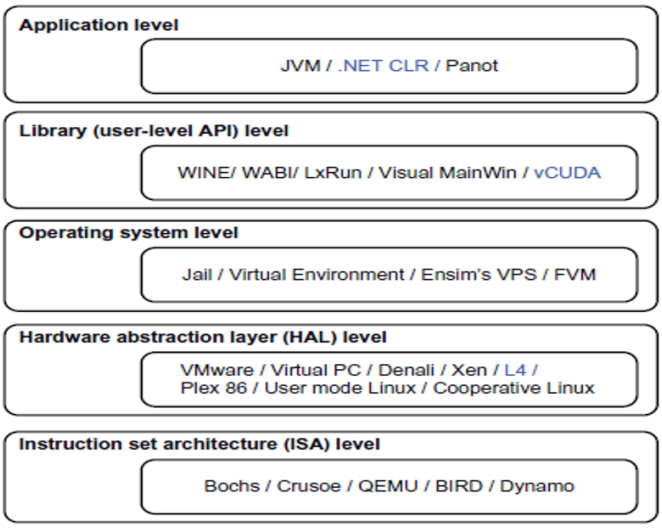


FIGURE 3.2
Virtualization ranging from hardware to applications in five abstraction levels.

Virtualization can be implemented at different layers of a computer system, each offering a different balance of performance, isolation, and flexibility.

1. Instruction Set Architecture (ISA) Level

Accumulator	Register (register-memory)	Register (load-store)
Load A Add B Store C	Load R1, A Add R1, B Store C, R1	Load R1,A Load R2, B Add R3, R1, R2 Store C, R3

- What is an Instruction Set? The part of a computer architecture related to programming, including native data types, instructions, registers, addressing modes, memory architecture, interrupt and exception handling, and external I/O. It's the language the processor understands (e.g., x86, MIPS).
- Virtualization: Transforming the physical architecture of the system's instruction set completely into software.

- This involves emulating a guest ISA using the host machine's ISA. The guest system's instructions are translated or interpreted by software so they can run on the host hardware.
- How it Works:
    - Guest systems issue instructions.
    - An emulator (software) receives these instructions.
    - The emulator translates the guest instructions into the native instruction set of the host machine.
    - The native instructions are executed on the host hardware.
- Methods:
    - Code Interpretation: Interprets and executes source instructions one by one. This is slow, as one source instruction may require many native instructions.
    - Dynamic Binary Translation: Translates blocks of frequently used ("hot") source instructions into target instructions for better efficiency.Instruction set emulation requires binary translation and optimization.
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- Example: Running MIPS binary code on an x86-based host machine. QEMU is a popular open-source emulator that can do this.
- Advantage: Excellent application flexibility. It allows legacy code written for one processor to run on entirely new hardware. Enables running an OS for one platform on another (e.g., x86 on SPARC).
- Enables the host system to adjust to a change in the architecture of the guest system, if accomplishing a task can be possible through the instructions that are available with the host system.
- Disadvantage: Poor performance. The interpretation and translation process introduces significant overhead, making it much slower than running code natively.
- An interpreter program interprets the source instructions to target instructions one by one. Therefore the system with the virtualization at ISA level shows a poor performance.
- QEMU is a generic and open source machine emulator and virtualizer.Run operating systems for any machine, on any supported architecture

2. Hardware Abstraction Level (HAL)

- What is it? Virtualization performed directly on top of the bare hardware. A hypervisor (VMM) manages the physical resources (CPU, memory, I/O devices) directly.
- How it Works: The hypervisor creates a virtual hardware environment for each VM. It must handle all interactions with the physical hardware, including privileged instructions from the guest OSes.
- Privileged Instructions: Instructions that can only run in kernel (supervisor) mode. If a VM attempts to execute a privileged instruction, it is trapped by the VMM, which then executes the instruction safely on behalf of the VM. This ensures isolation and system stability.
- Execution of privileged instructions needs full attention of CPU.If not managed properly by the VMM, will raise an exception resulting into system crash.
- Examples: VMware ESXi, Microsoft Hyper-V, Xen.
- Advantage: High performance (near-native) and good application isolation because each VM runs on its own virtualized hardware.
- Shortcoming: Very expensive and complex to implement.
- Note on Guest OS: If the physical and virtual OSes are the same, this can lead to duplication of effort in installation and administration.

3. Operating System Level

- What is it? Virtualization is implemented within a single operating system kernel to create multiple isolated user-space instances, often called containers, virtual environments (VEs), or jails.
- How it Works: It creates an abstraction layer between the OS and user applications. The host OS kernel is shared among all containers, making them extremely lightweight. Each container has its own isolated file system, processes, and network, but they all share the same kernel.
- Why use it? It overcomes the redundancy and time consumption issues of HAL virtualization, where each VM requires a full OS installation.
- Examples: Docker, LXC, FreeBSD Jails.
- Docker Analogy: A container is like a physical shipping container for software. It packages the application along with all its dependencies

and configuration, ensuring it runs consistently regardless of the environment.

- Containers vs. Virtual Machines:
    - VMs: Each VM runs a full, separate guest operating system. This leads to larger disk images and higher resource overhead. OS settings, dependencies, and patches are entangled within each VM.
    - Containers: Containers share the single kernel of the host operating system. A container image only needs the executable and its package dependencies. They run as isolated processes, are easy to manage individually, and are portable ("runs anywhere").
- Usage: Commonly used in virtual hosting environments to allocate hardware resources among a large number of mutually distrusting users.

4. Library Support Level

- What is it? Virtualization that controls the communication link between applications and the OS by intercepting Application Programming Interface (API) calls. It's also known as user-level ABI or API emulation.
- How it Works: Instead of virtualizing the whole OS, it creates an execution environment for alien (different OS) programs. It works by intercepting and remapping API calls from the application to the host system's libraries.
- Examples:
    - WINE (Wine Is Not an Emulator): Allows Windows applications to run on Unix-like operating systems by translating Windows API calls into POSIX (Portable Operating System Interface) calls on the fly.
    - vCUDA (Virtual Compute Unified Device Architecture): Allows applications inside VMs to access and leverage physical NVIDIA GPUs on the host machine for high-performance computing. It intercepts CUDA API calls from the guest and redirects them to the host.

- Advantage: Very low implementation effort compared to hardware-level virtualization.
- Shortcoming: Poor application flexibility and isolation.

5. User-Application Level

- What is it? Virtualization that sits as an application program on top of an OS and virtualizes an application or process. It's often called a process VM or High-Level Language (HLL) VM.
- How it Works: It exports an abstraction of a VM that can run programs compiled for that specific abstract machine. The classic example is a platform-independent programming environment.
- Examples:
    - Java Virtual Machine (JVM): Enables a computer to run Java programs. Java source code is compiled into platform-independent bytecode. The JVM interprets or compiles this bytecode into actions and OS calls for the specific platform it is running on.
    - .NET Common Language Runtime (CLR): Similar to the JVM, it runs bytecode (CIL - Common Intermediate Language) for .NET applications.
- Stack vs. Register VMs:
    - Stack-based (JVM, .NET CLI): Uses an execution stack for operations. Bytecode instructions load operands onto the stack, perform operations, and store the result back on the stack.
    - Register-based (Parrot VM): Uses virtual registers to hold operands, which can lead to faster execution but more complex code.
- Other Forms:
    - Application Isolation: Wrapping an application in a layer isolated from the host OS and other apps, making it easy to distribute and remove.
    - Application Streaming: Delivering application components to a user's computer on demand.

    - Example: LANDesk: An asset management system that remotely controls, inventories, and manages desktop computers.
- Advantage: Best application isolation.
- Shortcoming: Lower performance, low application flexibility, and high implementation complexity.

Relative Merits Summary

A comparison of the different virtualization levels based on performance, flexibility, isolation, and implementation cost.

Implementation Level	Performance	Application Flexibility	User Isolation	Implementation Complexity
ISA	Low	High	Low	High
HAL	High	Moderate	High	Very High
OS Level	High	Low	Moderate	Low
Library Level	Moderate	Moderate	Low	Low
Application Level	Low	Low	High	High

Overall, hardware-level (HAL) and OS-level support yield the highest performance, while ISA-level offers the best flexibility. Hardware and application levels are the most expensive to implement.

5. Virtualization Design Requirements

For a VMM to be effective, it must meet three classic design requirements:

1. Equivalence Requirement: A program running under the VMM should behave in essentially the same way as it would on equivalent physical hardware. The VM must be able to execute all the instructions and programs designed for the real machine.
2. Efficiency Requirement: The majority of guest instructions should be executed directly by the physical hardware, without intervention from the VMM. The goal is for the VM to be as efficient in performance as a real system. Virtualization is done to get efficient use of software without the physical hardware, so performance overhead must be minimal.
3. Resource Control Requirement: The VMM must have complete control over the system resources (processors, memory, I/O devices). It must be able to manage and allocate these resources and enforce isolation between VMs so that one VM cannot interfere with another.

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6. Virtualization Structures/Tools and Mechanisms

Hypervisors

- Definition: A hypervisor (or Virtual Machine Monitor - VMM) is a hardware virtualization technique that allows multiple guest

operating systems to run concurrently on a host machine. It converts physical devices into virtual resources for VMs to use.

- Types of Hypervisors:
    - Type 1 (Bare-Metal): Runs directly on the system hardware. It is the first layer on top of the hardware. (Examples: Microsoft Hyper-V, VMware ESXi, Xen).
    - Type 2 (Hosted): Runs as an application on top of an existing host operating system. The hypervisor is the second layer. (Examples: VMware Workstation, Oracle VirtualBox).

Hypervisor Architectures

- Micro-kernel Architecture: The hypervisor includes only the most basic and unchanging functions, like physical memory management and processor scheduling. Device drivers and other changeable components are kept outside the hypervisor. This results in a smaller, more secure code base. (Example: Microsoft Hyper-V, Xen).
- Monolithic Architecture: The hypervisor implements all functions, including device drivers, within itself. This can lead to a larger code base. (Example: VMware ESX).

The Xen Architecture

- Overview: Xen is a type 1, open-source, microkernel hypervisor originally developed at Cambridge University. It forms the base for many commercial and open-source virtualization products (e.g., Citrix XenServer).
- Core Technology: Primarily based on paravirtualization, though it now also supports full virtualization with hardware assistance. Its microkernel design means it does not include device drivers natively, keeping it small.
- Key Concepts - Domains:
    - Domain: A VM instance running on the Xen hypervisor.
    - Domain 0 (Dom0): A special, privileged guest OS (a "Management VM") that is the first to be loaded. It has direct

access to the hardware and manages all other guest domains. It hosts the drivers and a control stack to create, migrate, and terminate other VMs. If Dom0 is compromised, the entire system is at risk.
    - Domain U (DomU): All other, unprivileged guest domains running on the system. They do not have direct hardware access.
- Hypercalls: In paravirtualization, sensitive system calls in the guest OS are replaced with hypercalls. A hypercall is a software trap from a domain to the hypervisor, similar to how a system call (syscall) is a trap from an application to the OS kernel. This allows the guest OS to directly request services from the hypervisor.

Full Virtualization with Binary Translation

- Concept: A technique to virtualize architectures that are not naturally virtualizable (like x86). It combines direct execution and binary translation.
- How it Works:
    - The VMM runs in Ring 0 (most privileged), and the guest OS is run in Ring 1 (less privileged).
    - Non-critical instructions (user-level applications) are run directly on the hardware for efficiency.
    - The VMM scans the instruction stream of the guest OS. When it detects a critical instruction (privileged, control-sensitive, or behavior-sensitive), it is trapped.
    - The trapped instruction is replaced (translated) with an equivalent set of instructions that perform the same function but safely, without generating exceptions. This process is called binary translation.
    - The translated instructions are executed by the VMM on behalf of the guest.
- Popek and Goldberg Virtualization Requirements: A classic framework defining instructions for virtualization:
    - Privileged Instructions: Trap if in user mode, not if in system mode.

    - Control-Sensitive Instructions: Attempt to change the configuration of system resources.
    - Behavior-Sensitive Instructions: Behavior depends on the configuration of resources.
- Caching: To improve performance, translated "hot" instructions are stored in a code cache, so translation is not needed for subsequent executions. This increases performance but uses more memory.
- Advantage: The guest OS runs completely unmodified.
- Disadvantage: Binary translation introduces performance overhead, especially for I/O-intensive applications.
- Example: VMware Workstation and ESX use this technique.

Para-Virtualization with Compiler Support

- Concept: A technique where the guest operating system is modified to be virtualization-aware. It does not require binary translation.
- How it Works:
    - The guest OS kernel is modified to replace non-virtualizable instructions with hypercalls that communicate directly with the hypervisor.
    - This is often assisted by an "intelligent compiler."
- Advantage: Reduces virtualization overhead, leading to performance very close to native, often better than full virtualization.
- Disadvantage: Requires significant modifications to the guest OS, which can be a high maintenance cost. The modified OS can no longer run on hardware directly.
- Comparison:
    - Full Virtualization: Unmodified guest OS, binary translation for critical instructions.
    - Para-Virtualization: Modified guest OS, hypercalls for critical instructions.

VMware: A Case Study

- Overview: VMware is a pioneer in virtualization, offering a full range of solutions based primarily on full virtualization (combining direct execution and binary translation).
- End-user (Desktop) Virtualization: VMware Workstation (for Windows) and VMware Fusion (for Mac) are Type 2 hypervisors. They install a driver in the host OS that provides a privileged VMM and processes I/O requests by relaying them to the host OS.
- Server Virtualization:
    - VMware GSX Server: An early, hosted approach for servers.
    - VMware ESX / ESXi: Type 1, bare-metal hypervisors. ESXi is a smaller, more secure version of ESX. These are the core building blocks for VMware's data center products.
- Infrastructure and Cloud Solutions (vCloud Suite):
    - ESXi: The virtualization platform.
    - vSphere: The virtualization platform that pools ESXi hosts and provides core services like virtual compute, storage (Virtual SAN), and networking. It also offers application services like vMotion (live migration) and data recovery.
    - vCenter: Centralized management platform for controlling multiple vSphere installations.
    - vCloud: Turns virtualized data centers into an IaaS (Infrastructure-as-a-Service) cloud, allowing providers to offer on-demand, pay-per-use virtual environments.
    - vFabric: A PaaS (Platform-as-a-Service) solution with components for developing and running scalable web applications on the virtual infrastructure.
    - Zimbra: A SaaS (Software-as-a-Service) solution for email, collaboration, and office automation, hosted in the cloud.

Microsoft Hyper-V

- Overview: Microsoft's Type 1 hypervisor-based solution for server virtualization.
- Architecture - Partitions:
    - Partition: A completely isolated environment where an OS runs.

    - Parent Partition (Root Partition): Created first. It is a special VM instance (the original host OS) that has direct access to the hardware. It runs the virtualization stack, hosts all the hardware drivers, and manages the creation of child partitions.
    - Child Partitions: Guest VMs that run operating systems. They do not have direct hardware access and rely on the parent partition or hypervisor.
- Hyper-V Hypervisor Components:
    - Hypercalls Interface: The entry point for partitions to execute sensitive instructions via hypercalls (paravirtualization).
    - Memory Service Routines (MSRs): Controls memory access. It leverages hardware-assisted virtualization (I/O MMU) for fast device access.
    - Scheduler: Schedules virtual processors on physical processors.
    - Address Manager: Manages virtual network addresses for guest OSes.
    - Partition Manager: Handles partition creation, destruction, and configuration.
    - Advanced Programmable Interrupt Controller (APIC): Manages hardware interrupts (timer, I/O, exceptions). Each virtual processor has a Synthetic Interrupt Controller (SynIC) for this purpose. The hypervisor dispatches physical interrupts to these SynICs.
- Privilege Level: The Hyper-V hypervisor runs in Ring -1, a highly privileged mode enabled by modern hardware. This allows it to support both legacy and modern operating systems.

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7. Cloud Computing and Virtualization

The Cloud Reference Model

Cloud computing can be organized into a layered model covering the entire stack from hardware to applications.

1. Cloud Resources (Physical Infrastructure): The actual hardware, such as servers, storage, and network devices, often housed in large data centers.
2. Core Middleware (Virtualization Layer): Manages the physical infrastructure to provide a runtime environment for applications. It uses hypervisors to pool resources and expose them as collections of VMs. This layer enables customization, isolation, and QoS.
3. Cloud Programming Environments and Tools (User-Level Middleware): Offers developers a platform to build and deploy applications. This includes web-based interfaces, command-line tools, and frameworks for distributed programming.
4. Application Layer: The actual software services delivered to end-users.

Service Models (XaaS)

- IaaS (Infrastructure-as-a-Service): Provides on-demand access to IT infrastructure—servers, virtual machines, storage, networks, and operating systems. This corresponds to the combination of the Cloud Resources and Core Middleware layers.
    - *Provider Example:* Amazon EC2.
- PaaS (Platform-as-a-Service): Provides on-demand access to a complete, ready-to-use, cloud-hosted platform for developing, running, and managing applications. This corresponds to the User-Level Middleware layer, often bundled with the underlying IaaS.
    - *Provider Example:* Google App Engine, VMware vFabric.
- SaaS (Software-as-a-Service): Provides on-demand access to ready-to-use, cloud-hosted application software. This corresponds to the Application Layer.
    - *Provider Example:* Salesforce, VMware Zimbra.

Economics of the Cloud

The primary drivers of cloud computing are economies of scale and the simplicity of the pay-as-you-go model.

- Capital Costs (CapEx): One-time expenses for purchasing physical assets like servers and software licenses. These assets depreciate over time. Cloud computing helps reduce CapEx by shifting spending to operational expenses.
- Operational Costs (OpEx): Ongoing costs for running a business, like maintenance, electricity, cooling, and IT staff.
- Cloud Benefits:
    - Reduces capital costs for IT infrastructure.
    - Eliminates depreciation costs on IT assets.
    - Replaces software licensing with subscriptions.
    - Cuts maintenance and administrative IT costs.

Cloud Pricing Models

1. Tiered Pricing: Services are offered in fixed tiers with specific computing specifications and SLAs, each at a set price per unit of time (e.g., per hour). (Example: Amazon EC2's different instance types).
2. Per-Unit Pricing: Customers pay based on consumption of specific units, like data transfer or RAM/hour. This allows for more efficient configuration. (Example: GoGrid's RAM/hour model).
3. Subscription-based Pricing: Users pay a recurring fee for access to software or services, common in SaaS. (Example: Microsoft 365).

Open Challenges in Cloud Computing

1. Cloud Definition: While the NIST definition is widely accepted, alternative taxonomies exist (e.g., David Linthicum's 11 categories like Storage-as-a-Service, Database-as-a-Service) and academic ontologies (e.g., UCSB's five layers: applications, software environments, software infrastructure, software kernel, hardware).
2. Cloud Interoperability and Standards: Lack of standards leads to vendor lock-in, making it difficult and costly for customers to switch providers. Organizations like the Cloud Computing Interoperability Forum (CCIF) are working on this.
3. Scalability and Fault Tolerance: Designing cloud middleware that can scale seamlessly on demand and is resilient to failures is a major technical challenge.
4. Security, Trust, and Privacy: Major obstacles to adoption. Issues include data tampering, new threats from virtualization, trusting the provider, compliance with data privacy regulations, and complex chains of responsibility when a service uses multiple third-party providers.
5. Organizational Aspects: Cloud computing changes how IT is consumed and managed. Organizations must adapt to new billing models and answer questions about the new role of the IT department, how compliance will work with less control, and how end-users will perceive these new services.